MATHS MANTHAN
• TUITION CLASSES •
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Subject: Mathematics | **CBSE CLASS 10 – WEEKLY TEST 05** | **Time:** 45 Minutes
Class: Class X (CBSE) | **Chapter 5: Arithmetic Progressions** | **MM: 20**

Student Name: _____ **Roll No.:** _____ **Date:** _____

GENERAL INSTRUCTIONS:

1. All questions are compulsory. Internal choice is provided in Question 10(iii).
2. **Section A** comprises 4 MCQs and 1 Assertion-Reason question of 1 mark each.
3. **Section B** comprises 1 Very Short Answer question (2 marks) and 3 Short Answer questions (3 marks each).
4. **Section C** comprises 1 Applied Case Study question of 4 marks with internal choice.
5. Use of calculators is not permitted.

SECTION A: Multiple Choice & Assertion-Reason (5 × 1 = 5 Marks)

Q1. In an AP, if $a = 1$, $a_n = 20$, and $S_n = 399$, then n is: [1]

(A) 19 (B) 21 (C) 38 (D) 42

Q2. The sum of the first five multiples of 3 is: [1]

(A) 45 (B) 55 (C) 65 (D) 75

Q3. If the sum of first n terms of an AP is $S_n = 3n^2 + 5n$, then its common difference is: [1]

(A) 3 (B) 6 (C) 5 (D) 2

Q4. If $k, 2k-1, 2k+1$ are three consecutive terms of an AP, then the value of k is: [1]

(A) 2 (B) 3 (C) -3 (D) 1

Q5. **Assertion (A):** The sum of the first 10 natural numbers is 55.

Reason (R): The sum of the first n natural numbers is given by $S_n = \dfrac{n(n+1)}{2}$. [1]

(A) Both (A) and (R) are true; (R) explains (A).
(B) Both are true; (R) does not explain (A).
(C) (A) is true but (R) is false.
(D) (A) is false but (R) is true.

SECTION B: Short Answer Questions (11 Marks)

Q6. How many terms of the AP $24, 21, 18, \ldots$ must be taken so that their sum is 78? Explain the double answer. [2]

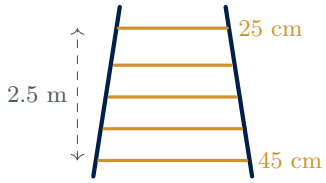
Q7. If the nth terms of two APs $9, 7, 5, \ldots$ and $24, 21, 18, \ldots$ are equal, find the value of n. [3]

Q8. For what value of p are $2p-1, 7, 3p$ consecutive terms of an AP? Also find the next term. [3]

Q9. Find the sum of the first 15 multiples of 8. [3]

SECTION C: Applied Case Study (1 × 4 = 4 Marks)

Q10. CASE STUDY: Step-Ladder Tapering Rungs Carpentry



A master carpenter constructs a traditional wooden step-ladder for domestic use. For ergonomic stability, the rungs of the ladder decrease uniformly in length from 45 cm at the bottom rung to 25 cm at the top rung.
The total vertical distance between the top rung and the bottom rung is 2.5 m (250 cm), and the rungs are placed at equal intervals of 25 cm center-to-center.

Based on the above mathematical scenario, answer the following questions:

(i) Determine the total number of rungs present in the step-ladder. [1]

(ii) Find the common difference in length between two consecutive rungs from bottom to top. [1]

(iii) Calculate the total length of wood required to manufacture all the rungs of the ladder. [2]

— OR —

Find the length of the middle rung of the ladder. [2]

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★★★ END OF QUESTION PAPER · ALL THE BEST ★★★